Logistic Regression

Recap on Naive Bayes

NB is a **generative model** which models $P(x, y)$

$$P(y \mid x) \propto P(y)P(x \mid y) = P(y)\prod_{i=1}^{d} P(x[i] \mid y)$$

Conditional independent assumption given label

Today

Logistic regression — a ***discriminative learning*** approach that directly models $P(y \mid x)$ for classification

Logistic Regression

Setting: binary classification $\mathscr{D} = \{x_i, y_i\}_{i=1}^n$, $(x_i, y_i) \sim P$,
$x_i \in \mathbb{R}^d, y_i \in \{-1, +1\}$

(Note, we always assume x contains a constant 1)

Logistic regression **directly models** $P(y \mid x)$

$$P(y \mid x) = \frac{1}{1 + \exp\left(-y(x^\top w^\star)\right)}$$

Logistic Regression

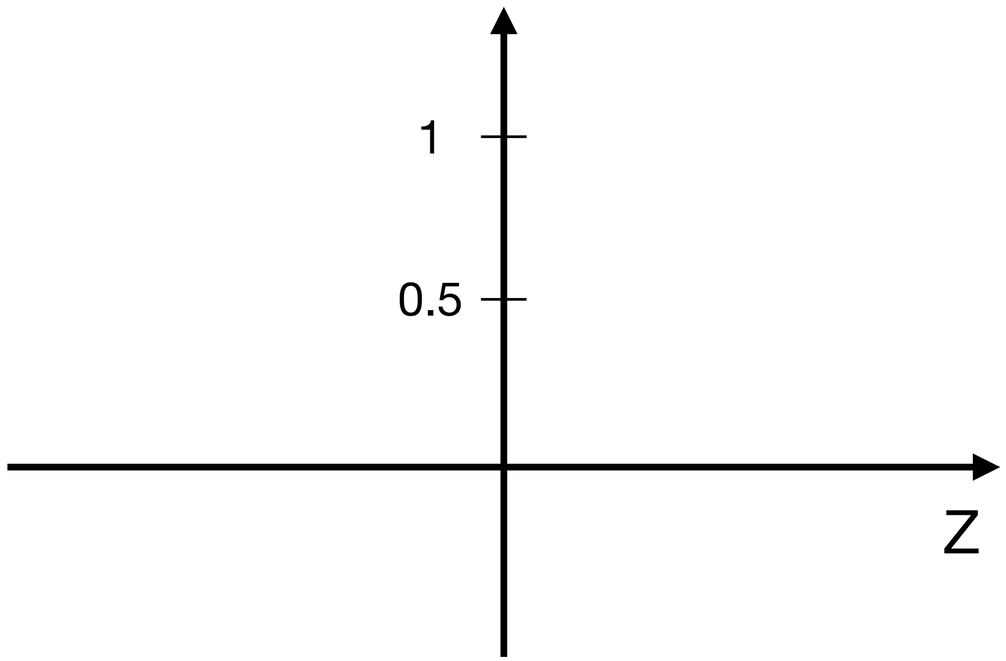
Logistic regression assumes:

$$P(y \mid x) = \frac{1}{1 + \exp\left(-y(x^\top w^\star)\right)}$$

The model assigns higher prob to

$$y = \text{sign}(x^\top w^\star)$$

Draw the Sigmoid function $1/(1 + \exp(-Z))$



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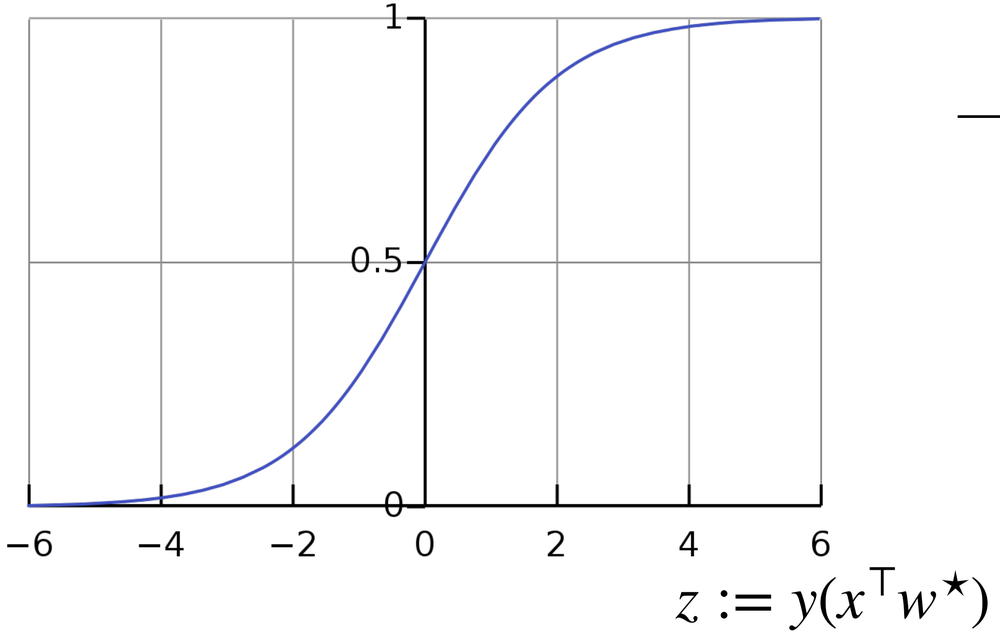
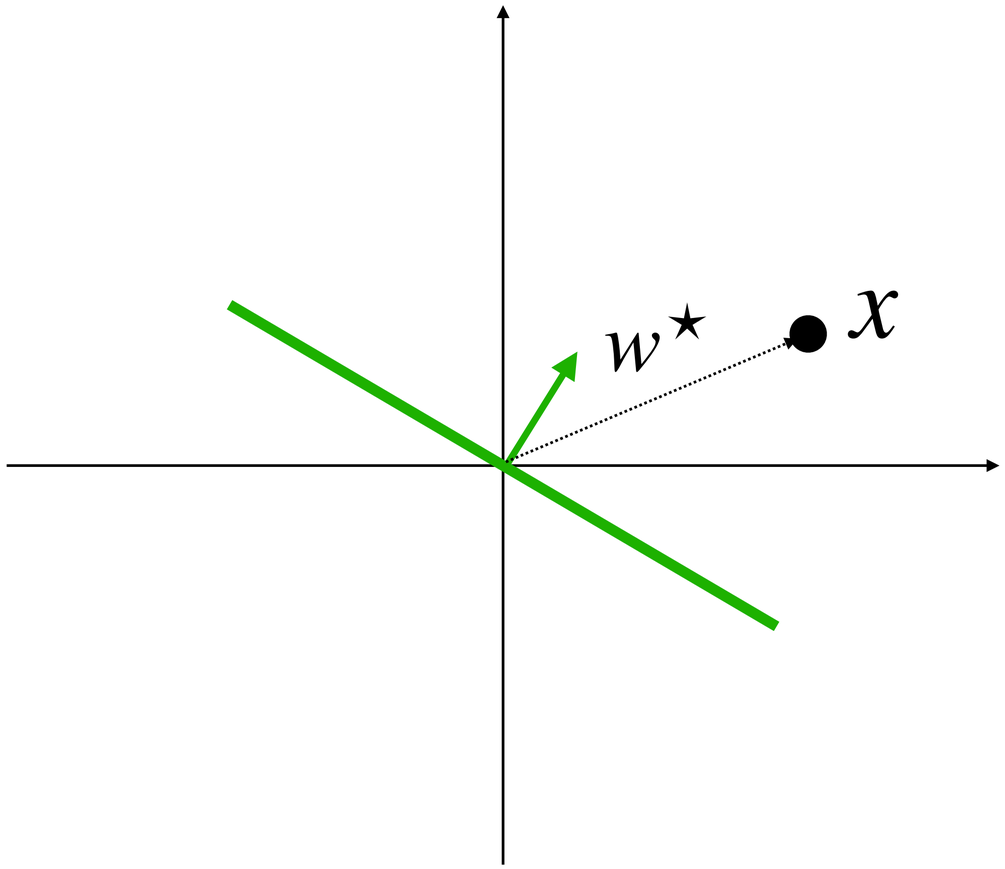




Learn via MLE

Recall we have data $\mathscr{D} = \{x_i, y_i\}_{i=1}^n$

$$\arg\max_w P(\mathscr{D} \mid w) = \arg\max_w P\left(\{y_i\}_{i=1}^n \mid \{x_i\}_{i=1}^n; w\right)$$

$$= \arg\max_w \prod_{i=1}^n P\left(y_i \mid x_i; w\right)$$

Plug in logistic assumption and add log:

$$\arg\max_w \sum_{i=1}^n -\ln\left[1 + \exp\left(-y_i(w^\top x_i)\right)\right]$$

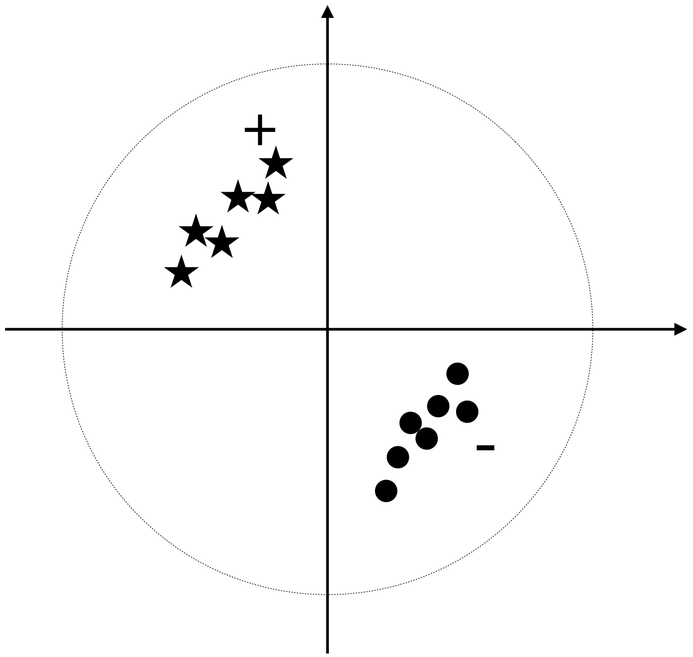
Learn via MLE

$$\hat{w}_{mle} := \arg\max_{w} \sum_{i=1}^{n} \ln \left[\frac{1}{1 + \exp\left(-y_i (w^\top x_i)\right)} \right]$$

Intuitively, $\hat{w}_{mle}$ tries to explain the label:

Q: for $y_i = +1$, what we should expect from $\hat{w}_{mle}^\top x_i$?

Q: for $y_i = -1$, what we should expect from $\hat{w}_{mle}^\top x_i$?



Learn via MAP

$$P(w \mid \mathscr{D}) \propto P(w)P(\mathscr{D} \mid w)$$

We use Gaussian prior, i.e., $P(w) = \mathscr{N}(0, \sigma^2 I)$

$$\arg\max_w \ln \left(P(w) \prod_{i=1}^{n} P(y_i \mid x_i, w) \right) = \arg\max_w \ln P(w) + \sum_{i=1}^{n} \ln P(y_i \mid x_i, w)$$

$$= \arg\min_w \left(\sum_{i=1}^{n} \ln \left(1 + \exp(-y_i(w^\top x_i))\right) + \frac{\|w\|_2^2}{2\sigma^2} \right)$$

Comparison to Navie Bayes

1. Logistic regression does not model $P(x \mid y)$

2. Gaussian NB leads a linear classifier in the form of
$$P(y \mid x) = 1/(1 + \exp(w^\top x))$$

Gaussian NB is a special case of logistic regression

https://mlpocket.com/ml/supervised/logistic-regression

Let's summarize by applying GD to logistic regression

Recall the objective for LR:

$$\min_{w} \sum_{i=1}^{n} \ln\left[1 + \exp\left(-y_i(w^\top x_i)\right)\right] + \lambda \|w\|_2^2$$

Initialize $w^0 \in \mathbb{R}^d$

Iterate until convergence:

1. Compute gradient $g^t = \sum_i \frac{\exp(-y_i x_i^\top w^t)(-y_i x_i)}{1 + \exp(-y_i x_i^\top w^t)} + 2\lambda w^t$
2. Update (GD): $w^{t+1} = w^t - \eta g^t$